

Complete Firebase Cost Analysis

PG Owner SaaS

Billing Mechanics · Data Architecture · SaaS Scaling · Pricing Strategy

■ **75–130**

Monthly cost
200-bed PG

90%+

Profit margin
at ■15/bed

10+ yrs

Before DB
storage cost

15 PGs

Scale before
bandwidth bill

PART 1: How Firebase Billing Actually Works (Your Core Questions Answered)

QUESTION 1: IS STORAGE MONTHLY OR ONE-TIME?

Monthly recurring — every single month. Think of it like rent for a storage unit. Firebase looks at how much data you have stored at the end of each month and charges you for it. There is no concept of "I already paid for this data."

Example: You store 3 GB in January → pay for 3 GB. Same data still there in February → pay for 3 GB again. Delete it in March → pay \$0.

For Realtime Database on Blaze plan:

- First 5 GB stored: **FREE every month**
- Beyond 5 GB: **\$5 per GB per month** (recurring)

For Cloud Storage on Blaze plan:

- First 5 GB stored: **FREE every month**
- Beyond 5 GB: **\$0.026 per GB per month** (recurring)

QUESTION 2: WHAT COUNTS AS "DOWNLOAD" / BANDWIDTH?

Every byte of data read by any client = download. This includes:

- Your app opening and loading dashboard data
- Realtime listeners receiving updates when data changes
- Any `.get()`, `.onValue`, `.once()` call
- Residents opening the resident app and loading their info
- Images/files loaded from Cloud Storage

Service	Free/Month	After Free
Realtime DB bandwidth	~30 GB/month (1 GB/day)	\$1.00 per GB
Cloud Storage bandwidth	~30 GB/month (1 GB/day)	\$0.12 per GB

QUESTION 3: PHONE OTP (FIREBASE PHONE AUTH) PRICING

This is the most confusing and highest actual cost for your app.

On the **Spark (free) plan**: 10 OTP SMS per day maximum — hard cap, not scalable for production.

On the **Blaze plan**:

- First ~10 SMS/day free (tiny)
- Beyond that: charged per SMS by destination country
- India rate: approximately **₹0.23–₹0.42 per OTP** (\$0.0028–\$0.005)

Firebase does NOT publish a clear table in their main docs — the per-country rates are only visible after you sign up in the Firebase console under Billing. The pricing agent confirmed India is approximately **\$0.003–\$0.005 per verification**.

There is no monthly free bundle — just 10/day free, then pay per OTP.

QUESTION 4: CLOUD FUNCTIONS

Dimension	Free/Month	After Free
Invocations	2,000,000	\$0.40/million
Compute (GB-seconds)	400,000	\$0.0000025 each
External network egress	5 GB	\$0.12/GB

Your app has 1 Cloud Function (push notifications). At even 10,000 notifications/month, cost = **\$0 — completely in free tier**.

PART 2: Your App's Data Analysis (From Codebase)

DATABASE STRUCTURE

Your Firebase has these main paths:

/Owners/{uid}	owner/staff accounts
/Pgs/{pgId}	PG metadata
/PgsInfo/{pgId}/floorInfo	ALL rooms, beds, allocation history
/PgsInfo/{pgId}/payments/	master + by_month + by_resident indices
/Residents/{pgId}/	all resident records

/PgsInfo/{pgId}/notifications/	notifications + per-user read states
/PgsInfo/{pgId}/expenses/	all expenses
/PgsInfo/{pgId}/serviceRequests/	service requests
/login_ref/ and /owner_login_ref/	phone→user mapping

HOW MUCH DATA IS LOADED ON APP OPEN

Your app opens **16–18 simultaneous real-time listeners** on every dashboard load. No pagination exists — ALL data is loaded into memory.

What Gets Loaded	Data Size (50-bed PG)	Data Size (200-bed PG)
Owner/staff records	5 KB	5 KB
PG metadata + config	4 KB	4 KB
Floor/room/bed tree	120 KB	500 KB
Resident records	100 KB	400 KB
Payments (current month)	60 KB	250 KB
Service requests	25 KB	100 KB
Notifications (+ user states)	100 KB	600 KB
Expenses	60 KB	250 KB
Other (machines, menu, etc.)	10 KB	25 KB
Total per session open	~490 KB	~2.1 MB

DATA SIZE PER RECORD

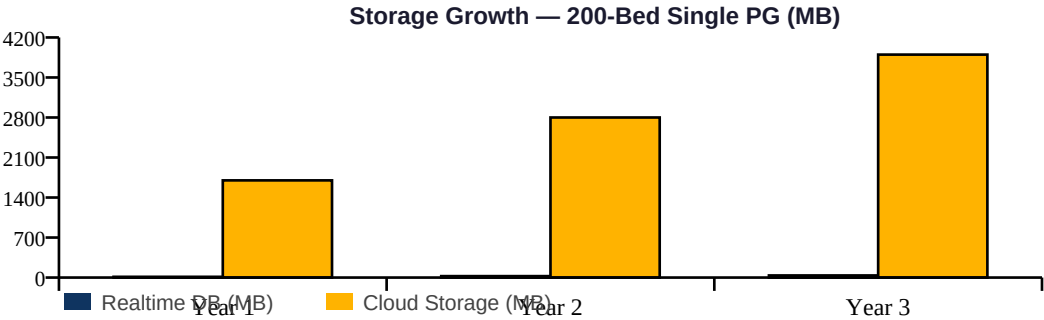
Model	Size Per Record	Notes
Resident	1.5–2 KB	Includes 3 embedded documents
Payment	1.5–3 KB	Higher with attachments
Room config	5–8 KB	Includes full allocation history
Notification	1–3 KB base + 60 bytes per user who saw it	Scales badly with users
Expense	1–1.5 KB	With image URLs
Service request	~1 KB	
Owner/staff	~900 bytes	

PART 3: Storage Growth Projections — Single 200-Bed PG

REALTIME DATABASE STORAGE GROWTH

Data Category	Year 1	Year 2	Year 3
Residents (200 active)	0.4 MB	0.4 MB	0.4 MB
Rooms + beds + allocation history	0.8 MB	1.2 MB	1.6 MB
Payments (200/month)	4.8 MB	9.6 MB	14.4 MB
Expenses (100/month)	1.4 MB	2.9 MB	4.3 MB
Notifications + user states	4.8 MB	9.6 MB	14.4 MB
Service requests	0.3 MB	0.6 MB	0.9 MB
Daily menus	0.3 MB	0.6 MB	0.9 MB
Misc (config, machines, etc.)	0.1 MB	0.1 MB	0.1 MB
Total DB size	~12.9 MB	~25 MB	~37 MB

DB storage cost: \$0 for at least 10+ years per PG. The 5 GB free tier is enormous compared to your text-only data.



CLOUD STORAGE (IMAGES/FILES) GROWTH

File Type	Per PG (Year 1)	Per PG (Year 2)	Per PG (Year 3)
Resident ID documents (3 docs × 200 residents × ~1 MB)	600 MB	600 MB	600 MB
PG profile image	2 MB	2 MB	2 MB
Expense bill photos (100/month × 0.5 MB)	600 MB	600 MB	600 MB
Service request images (50/month × 0.8 MB)	480 MB	480 MB	480 MB
Payment proofs	~0 (auto-deleted)	~0	~0
Total Cloud Storage	~1.7 GB	~2.8 GB	~3.9 GB

Cloud Storage cost: \$0 for first 3–4 years per PG (5 GB free tier covers it). Year 4 onwards, maybe \$0.05–\$0.10/month.

PART 4: Monthly Cost Breakdown — 200-Bed Single PG

BANDWIDTH (THE REAL VARIABLE COST)

Who Reads	Sessions/Day	Data/Session	Monthly Total
Owner (dashboard opens)	3	2.1 MB	189 MB
Staff (5 members)	2 each	1.5 MB	450 MB
Residents (200, 50% active/day)	100 sessions	0.5 MB	1,500 MB
Real-time push updates (all listeners)	continuous	—	~500 MB/month
Total bandwidth/month			~2.6 GB/month

Blaze free tier: 30 GB/month → you're well within free tier for a single PG.

PHONE OTP COST (MAIN RECURRING COST)

User Type	OTPs/Month	Cost @ \$0.33/OTP
Resident logins (200 × 1/month avg)	200	\$66
New resident registrations (~10/month)	10	\$3.30
Staff logins (5 × 3/month)	15	\$5
Owner	2	\$0.66
Total	~227 OTPs	~\$75/month (\$0.90)

If residents login more frequently (2x/month): ~\$130/month (\$1.55)

SINGLE 200-BED PG MONTHLY COST SUMMARY

Cost Item	Monthly Cost
Realtime Database storage	\$0
Realtime Database bandwidth	\$0
Cloud Storage (files)	\$0
Cloud Storage downloads	\$0
Phone OTP SMS	\$75–\$130
Cloud Functions	\$0
Total Firebase cost	\$75–\$130/month (\$0.90–\$1.55)

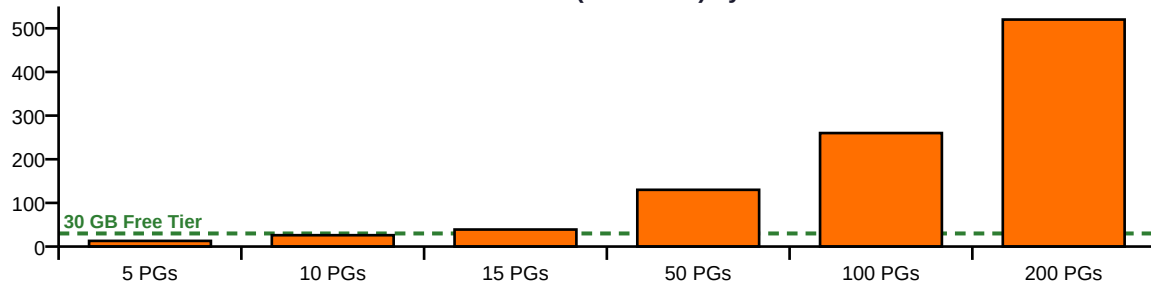
PART 5: SaaS Scale — What Happens With Multiple PGs

FREE TIER IS SHARED ACROSS YOUR ENTIRE PROJECT

When you have 100 PGs all using one Firebase project, free tiers are pooled together.

PGs	DB Bandwidth/Month	Free	Billable	Cost
5 PGs	13 GB	30 GB free	0	\$0
10 PGs	26 GB	30 GB free	0	\$0
15 PGs	39 GB	30 GB free	9 GB	\$9/month
50 PGs	130 GB	30 GB free	100 GB	\$100/month
100 PGs	260 GB	30 GB free	230 GB	\$230/month
200 PGs	520 GB	30 GB free	490 GB	\$490/month

DB Bandwidth (GB/Month) by Scale



PHONE OTP COSTS SCALE LINEARLY

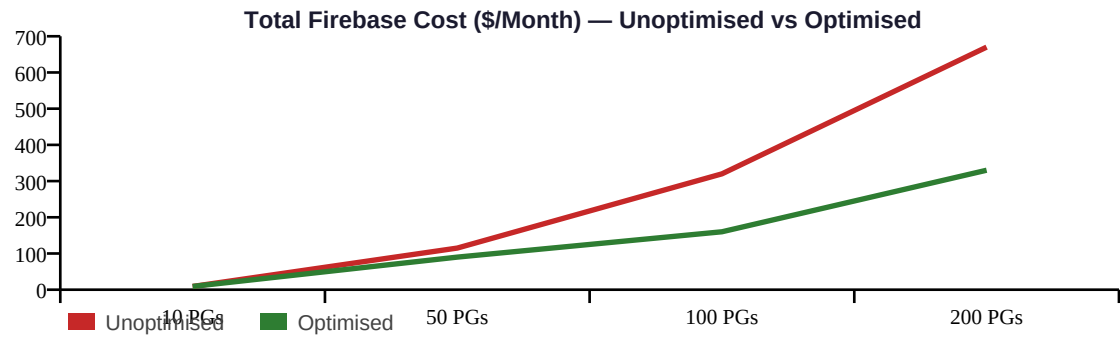
PGs	OTPs/Month	Cost (India)
10 PGs	2,270	₹750 (~\$9)
50 PGs	11,350	₹3,745 (~\$45)
100 PGs	22,700	₹7,490 (~\$90)
200 PGs	45,400	₹14,980 (~\$180)

TOTAL FIREBASE COST PROJECTION (SAAS)

Scale	DB Bandwidth	Phone OTPs	Total/Month	Per PG
10 PGs	\$0	\$9	\$9/month	\$0.90
50 PGs	\$70	\$45	\$115/month	\$2.30
100 PGs	\$230	\$90	\$320/month	\$3.20
200 PGs	\$490	\$180	\$670/month	\$3.35

After fixing pagination/no-load-all issues, bandwidth drops ~70%:

Scale (optimized)	DB Bandwidth	Phone OTPs	Total/Month	Per PG
100 PGs	\$70	\$90	\$160/month	\$1.60
200 PGs	\$150	\$180	\$330/month	\$1.65



PART 6: What to Charge Per Bed

YOUR ACTUAL COST PER BED (200-BED PG AT 100-PG SCALE)

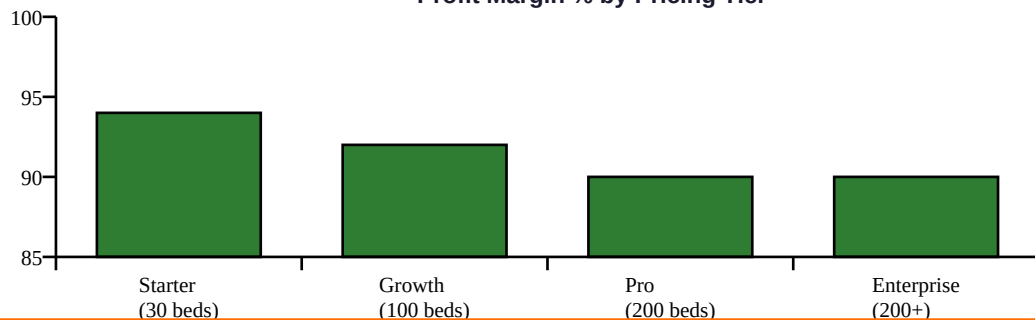
- Firebase cost per PG: ~₹240–₹270/month (\$3.20)
- Cost per bed: ₹1.2–₹1.35/bed/month

RECOMMENDED PRICING MODEL

Tiered by bed count (monthly subscription):

Tier	Beds	Suggested Price	Firestore Cost	Margin
Starter	Up to 30 beds	₹499/month	₹240;30	94%
Growth	31–100 beds	₹999/month	₹240;80	92%
Pro	101–200 beds	₹1,999/month	₹240;200	90%
Enterprise	200+ beds	₹3,499/month	₹240;350	90%

Profit Margin % by Pricing Tier



ALTERNATIVE: PER-BED PRICING

- ₹15–₹20/bed/month is very reasonable for the Indian market
- $200 \text{ beds} \times ₹15 = ₹3,000/\text{month}$ → Firestore cost ₹240–₹270 → **91% margin**

BREAK-EVEN ANALYSIS

You break even if you charge more than your Firestore cost. Since Firestore costs ₹1–₹1.5/bed/month, literally any charge above ₹2/bed/month is profitable. Even ₹5/bed gives you **70%+ margin**.

PART 7: Critical Issues That Will Hurt You at Scale

The codebase analysis found **10 inefficiencies**. The top 3 that directly affect your Firebase bill:

1. No Pagination on Payments

HIGH IMPACT

- **Problem:** When viewing "All Months", the app loads every single payment record ever created.
- 200-bed PG, 2 years old: $4,800 \text{ payment records} \times 2 \text{ KB} = \mathbf{9.6 \text{ MB per "All Months" view}}$
- **Fix:** Add `.limitToFirst(50)` with pagination — reduces this 95%

2. Notifications User States Blow Up

HIGH IMPACT

- **Problem:** Every notification stores a state entry for every user who saw it.
- $100 \text{ notifications} \times 200 \text{ users} \times 60 \text{ bytes} = \mathbf{1.2 \text{ MB just in read-states}}$
- After 1 year: $600 \text{ notifications} \times 200 \text{ users} = \mathbf{7.2 \text{ MB of user-state data}}$
- This downloads on **EVERY** dashboard open
- **Fix:** Move user states to a separate path, lazy-load them

3. Full Floor Tree Always Loaded

MEDIUM IMPACT

- **Problem:** Every room, every bed, every allocation from the beginning of time loads on dashboard open.
- A mature 200-bed PG with 5 allocation records per bed: $200 \times 5 \times 200 \text{ bytes} = \mathbf{200 \text{ KB just in history}}$
- Grows forever
- **Fix:** Truncate allocation history in the tree, archive old records

Summary — Is Firebase Profitable for Your SaaS?

Is storage monthly recurring?	Yes, every month
Will a 200-bed PG's DB data cost you?	No — 12 MB after year 1 vs 5 GB free
What is the main cost?	Phone OTP SMS (~■75–■130/PG/month)
Firebase cost per 200-bed PG	■75–■270/month (\$1–\$3.25)
Recommended charge per bed	■15–■20/bed/month
Margin at ■15/bed on 200 beds	~90%
Is it profitable?	Extremely — the business model works
Main risk	DB bandwidth at 50+ PGs if you don't fix pagination
Priority fix	Implement pagination on payments and lazy-load notifications

■ PRIORITY ACTION

The Firebase cost is **tiny** compared to what you can charge. Your biggest financial risk is not Firebase costs — it's the **bandwidth bill** if **your app keeps loading all data without pagination when you reach 15+ PGs**. Fix that before you onboard more customers.